

A Machine Learning Approach to Predicting Imminent Batting Collapse in T20 Cricket Using Ball-by-Ball Data

Abstract

Batting collapses are important in T20 cricket as they can alter the situation rapidly, but the majority cricket prediction research has been carried out on the final result, winner, or win probability. The authors of this study suggest the ball-by-ball ML model to predict imminent batting collapse based on T20 match data. A collapse happens when the batting side loses two wickets or more in a 12 legal balls in which they score 15 runs or less. This was created from Cricsheet ball-by-ball data, and covered 1,030,579 legal-ball states in 4,801 T20 matches. Match context, score state, recent scoring momentum, wicket history, batter and bowler data, and partnership data were extracted for leak-free features. The models considered were Logistic Regression, CatBoost, XGBoost, Temporal Convolutional Network and LightGBM. Of these, XGBoost gave the highest overall performance with 80.38% accuracy, 0.7079 ROC-AUC, 0.3148 PR-AUC and 0.3429 F1-score in the independent test set. The results demonstrate that it is possible to model the risk of a batsman collapsing within a short term of the match using ball-level match states, which offers a promising direction for early-warning modeling in the real-time cricket analytics and decision-making landscape.

CCS Concepts

• **Computing methodologies** → **Machine learning; Supervised learning; Ensemble methods**; • **Information systems** → *Data mining*; • **Applied computing** → Computers in other domains.

Keywords

T20 Cricket, Batting Collapse Prediction, XGBoost, Ball-by-Ball Analysis, Machine Learning.

1 Introduction

Twenty20 (T20) is a shortened form of cricket, where just a few deliveries can make a big difference to an innings. Partnerships are constantly re-formed and batters are constantly under pressure to score when wickets keep falling. When combined with a lull in the run scoring, the batting team may lose both resources and momentum in a short amount of time. This sudden decline is significant and in English domestic T20 cricket, it has been found that wicket preservation and forming huge partnerships are key factors for success in the format [11].

With the growing number of ball-by-ball datasets, fine-temporal predictive modeling has become possible. Most studies in the area of cricket analytics, however, tend to be based on end-of-match results, like predicting the match-winner or estimating the final score or predicting win-probabilities. For instance, the recent surge of T20 machine learning research has focused on predicting the result of a match given the match-level or team-level data, suggesting that this is one of the key research problems in cricket [6]. In-play prediction techniques are an extension of this where predictions are updated during the match and dynamic T20 winner prediction

has been explored with intermediate match states and relative team strength [14].

In tactical decision making, however, analysts need short horizon risk signals, ones that are based on the current situation of the innings as opposed to a one shot probability. A principled batter-capacity representation would be based on the idea of resources remaining with the expected scoring potential being a function of both balls remaining and wickets in hand. A relationship between these has been studied in T20 cricket, using adaptations to the Duckworth–Lewis resource model [4]. Similarly, contextual stress has been operationalized as a pressure signal in T20 run chases, supporting the broader idea that match states can be assigned to probabilistic risk measures [10].

The aim of this study is to suggest a ball-by-ball machine learning solution to forecasting a batter's impending collapse in T20 cricket. The problem is defined as a binary classification problem with short horizon (12 legal deliveries), where the task is to decide whether the batting side will "collapse" or not. Five models are tested in an identical experimental setup using Cricsheet data-derived features such as match-state, scoring, momentum, wicket, batter, bowler and partnership using leakage-free features. XGBoost shows the best test performance and is the model we propose to predict early batting-collapse risk.

This research study is summarized in the following key contributions:

- (1) **Ball-by-ball collapse prediction:** This study presents the imminent crashing of the batsman into a ball-by-ball binary classification problem in a future window of 12 legal deliveries in T20 cricket.
- (2) **Leakage-free feature design:** The features used in the current and past matches are designed based on a score, wickets, momentum of past matches, battery, bowler, and the partnership information; while the variables used in the future window are not added to the model input as this may cause data leakage.
- (3) **Comparative model evaluation:** Five machine learning models (Logistic Regression, CatBoost, XGBoost, Temporal Convolutional Network (TCN), and LightGBM) are evaluated in the same experimental setting and XGBoost is selected as the proposed model based on their test performance.

2 Related Works

A T20 batting collapse can turn the tide of the game and the course of a match in an instant. This section reviews the related cricket prediction studies thematically and summarizes them in table 1, and this section reviews them thematically.

The attention of the cricket analytics has shifted now to predicting scores, predicting the match outcomes, forecasting probability of victory, and analysing player performances. Current literature reveals that most of the studies focus on general prediction tasks like final result, team performance, and score prediction, whereas

predicting short-term events within an innings is relatively under-researched [15].

Several studies have used ball-by-ball and match-state data for predictive modelling. In the field of big-data cricket analytics, it has been found that the features like overs, wickets, recent runs, and recent wickets are helpful to predict the score and outcome of a cricket match [3]. Similar to in-play forecasting studies also reveal that the accuracy of predictions increases with increased match context information during the game [1, 2]. But these works primarily are about predicting the final score or who will be the winner, not about short-term batting failures.

Predicting using ball-by-ball data has also been investigated for T20 cricket. In-game win-probability models have been demonstrated to be able to be updated over the course of the match [8], and dangerous-ball analysis has been used to demonstrate how wickets and extras can be crucial for the outcome of a T20 match [9]. These studies share many of the same concepts as collapse behaviour studies, but they are aimed at the eventual goal of win probability outcome.

Another Markov-based models have been used to predict batsman performance from consecutive-ball behaviour [12]. In limited-overs run chasing, the use of CUSUM-based monitoring has been proposed to identify any changes in momentum [5]. These studies indicate that the ball history is valuable but they don't define the problem of batting collapse as a binary classification problem in the near future.

The tree-based and boosting methods, such as XGBoost, have demonstrated promising potential for cricket prediction as they can handle nonlinear relationships between match variables, which are inherent in cricket data [13]. Overall, previous studies tend to forecast the final score, player performance or win/loss probability, or the winner. This is in contrast to the focus of this study on early detection of imminent batting collapse after each legal ball in T20 cricket.

In general, this research is considered as a binary classification problem on a per-ball basis for the T20 batting-collapse prediction. The idea is to calculate the probability of the batting team losing at least two wickets in the next 12 legal balls at every given state of the game after the ball has been bowled, when the batter is in a legal position. The paper introduces the concept of match context, score state, recent scoring patterns, wicket history, and previous ball sequence features, which enable the study to go beyond traditional final outcome prediction to detecting early signs of collapse.

3 Methodology

This paper addresses the task of imminent batting collapse prediction in T20 cricket by formulating it as a binary classification problem on ball-by-ball basis with leakage-free features derived from the match state information from the cricsheet data. The goal is to estimate the risk of collapse in the next 12 legal deliveries as described in Figure 1.

3.1 Dataset

The dataset was created using the publicly available ball-by-ball cricket data from a structured repository of cricket match data, Cricsheet [7], Cricsheet also offers machine-readable match data,

such as detailed ball-by-ball data, making it an ideal tool for cricket analysis and prediction research. The raw match files were cleaned and matched and arranged into a modified csv data set for the purpose of this study. The match state is captured in each row, which includes contextual, score-based, recent performance, batter, bowler, and innings-level information for a legal delivery. This initial processed CSV had 1,085,959 rows and 71 columns. The final modeling dataset comprised of 1,030,579 records for 4,801 T20 matches and 9,612 innings, following the rejection of observations made when the ball was outside the laws of the game. Following the elimination of observations made when the ball was outside the laws of the game, the final modeling dataset had 1,030,579 observations for 4,801 T20 matches and 9,612 innings. This data was used to build and test models to accurately predict if a batsman would collapse in the next 12 legal deliveries.

3.2 Collapse Label Construction

The prediction task was formulated as a binary classification problem. Since batting collapse is not directly available in the raw ball-by-ball data, the target label was constructed using a fixed future window of 12 legal deliveries. For each ball state, a collapse was defined as losing at least two wickets while scoring no more than 15 runs in the next 12 legal balls.

Formally, the target label y_i for ball state i was defined as:

$$y_i = \begin{cases} 1, & \text{if } W_i^{(12)} \geq 2 \text{ and } R_i^{(12)} \leq 15, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where $W_i^{(12)}$ denotes the wickets lost and $R_i^{(12)}$ denotes the runs scored in the next 12 legal deliveries. In the implemented dataset, these correspond to `future_wkts_next_12` and `future_runs_next_12`. These future-window variables were used only to construct the target label and were not included as input features, ensuring a leakage-free prediction setting.

3.3 Leakage-Free Feature Engineering

To prevent data leakage, only current and past ball-state information was used as input features. Future-window variables used for label construction, including `future_wkts_next_12` and `future_runs_next_12`, were excluded from the model inputs.

The final feature set included 45 leakage-free variables: 40 numerical and 5 categorical features. The categorical features were `phase`, `batting_team`, `striker_id`, `non_striker_id`, and `bowler_id`. The main feature groups used in the final model are summarized in Table 2.

3.4 Data Splitting, Preprocessing, and Class Imbalance Handling

To avoid match-level leakage, the dataset was split by `match_id` into training, validation, and test sets using a 70:15:15 ratio. Missing values were handled using training-set statistics only; non-applicable chase features were filled with `-1`, numerical values with training medians, and categorical features were converted to categorical format.

The dataset was imbalanced, with 87.59% no-collapse and 12.41% collapse samples. Therefore, class weighting was applied where

Table 1: Summary of Existing Cricket Prediction Studies and Research Gaps

Ref.	Year	Dataset	Method	Result	Limitation
[8]	2024	IPL/T20 ball-by-ball data	Bayesian dynamic logistic regression	Improved Brier score and accuracy over baseline	Predicts win probability, not batting collapse
[9]	2025	2492 T20I matches	LR, MLP, Decision Tree	Best accuracy: 76.9%	Uses limited features; predicts match outcome
[5]	2019	1180 ODI and 537 T20I matches	Optimized CUSUM charts	76–84% accuracy for T20	Focuses only on run chases
[2]	2016	606 ODI matches	Dynamic logistic regression	Over 72% first-innings and 82% second-innings accuracy after 10 overs	ODI-based; over-level prediction
[1]	2024	1359 ODI matches	NB, DT, RF, LR, k-NN, ANN	63–68% first innings; 72–76% second innings	Predicts winner, not short-term collapse
[12]	2023	Cricinfo ball-by-ball data	HMM and Markov chains	70.16–72.84% accuracy	Predicts batsman score, not team collapse
[15]	2022	ML cricket studies from 2001–2021	Systematic review	Not applicable	No predictive model developed
[13]	2024	T20 Kaggle/Cricsheet data	XGBoost and regression models	MAE = 1.73	Score prediction, not collapse prediction
[3]	2021	ODI Cricsheet data, 350,899 records	Regression and Spark ML	95% test accuracy; RMSE = 30.2	ODI score prediction; not T20 collapse-focused

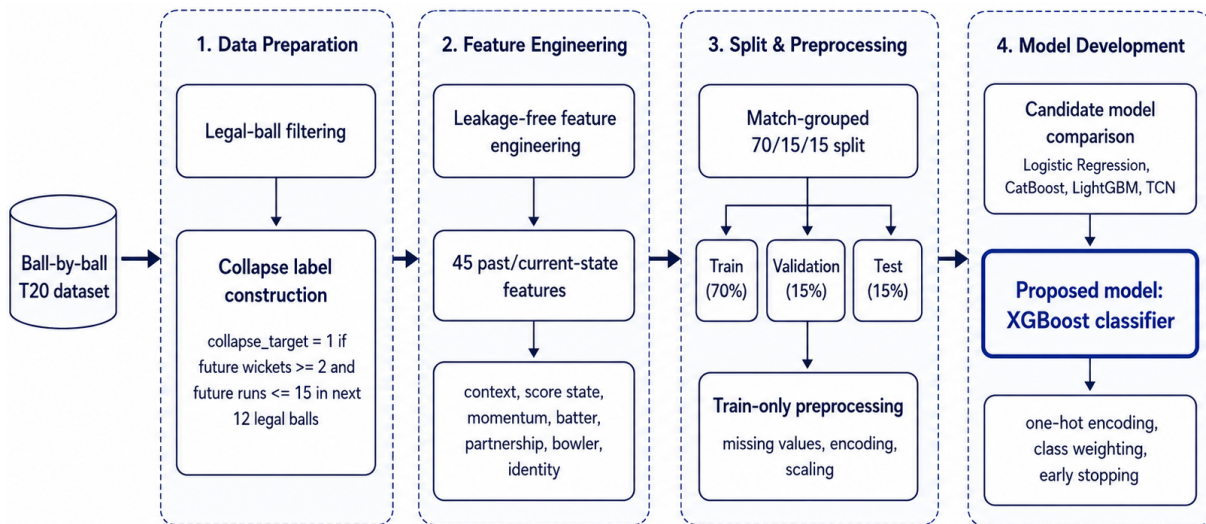


Figure 1: Methodological workflow for ball-by-ball T20 batting collapse prediction.

Table 2: Summary of leakage-free feature groups.

Feature Group	Examples
Match context	Innings, over, phase, batting team
Score state	Runs, wickets, run rate
Recent momentum	Recent runs, wickets, dot balls, boundaries
Batter information	Striker and non-striker state
Bowler information	Bowler identity and bowling state
Partnership status	Partnership runs and balls
Identity features	Team and player identifiers

supported, including `scale_pos_weight` for XGBoost and positive-class weighting for TCN, to reduce majority-class bias.

3.5 Candidate Model Development

Five candidate models were developed under the same train-validation-test setting: Logistic Regression, CatBoost, XGBoost, TCN, and LightGBM. Logistic Regression was used as a linear baseline, while CatBoost, XGBoost, and LightGBM represented tree-based gradient boosting models trained on the leak-free tabular feature set.

To capture short-term batting momentum, a TCN was also developed using the previous 12 legal deliveries for each ball state. This sequence input included recent runs, extras, dot balls, boundaries, wickets, wicket count, over, and legal ball position. All models were fairly evaluated, and The model that performed best on the

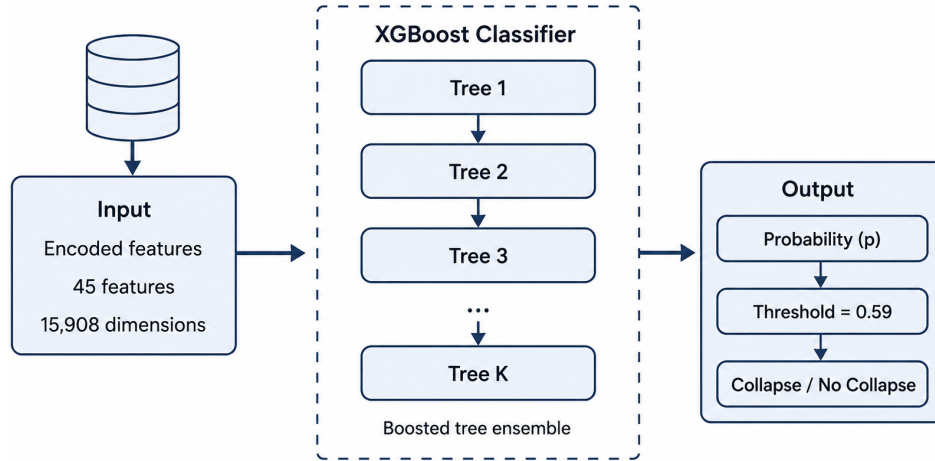


Figure 2: Proposed XGBoost architecture for imminent batting collapse prediction.

independent test set was selected as the proposed batting collapse prediction model.

3.6 Model Analysis and Evaluation

The five candidate models were evaluated in the same experiment under the same settings using accuracy, ROC-AUC, PR-AUC, F1-score, precision, recall and Brier Score. This subsection reports the performance of the baseline and alternative models and the proposed model, which is the best-performing model of XGBoost, is described separately.

- (1) **Logistic Regression:** Logistic Regression served as the linear baseline and achieved 75.64% accuracy, 0.6506 ROC-AUC, 0.2096 PR-AUC, and 0.2978 F1-score at the optimal threshold of 0.57. Its relatively low PR-AUC indicates limited effectiveness in detecting collapse cases under class imbalance.
- (2) **CatBoost:** CatBoost obtained 74.85% accuracy, 0.6634 ROC-AUC, 0.2491 PR-AUC, and 0.2954 F1-score at a threshold of 0.64. Although PR-AUC improved compared to Logistic Regression, its overall collapse-detection performance remained below the best model.
- (3) **Temporal Convolutional Network (TCN):** The Temporal Convolutional Network (TCN): The TCN used features of the last 12 legal deliveries. At a threshold of 0.52, it achieved 67.75% accuracy, 0.6209 ROC-AUC, 0.1888 PR-AUC, and 0.2672 F1-score. It has a high recall of 0.4793 which means that it is more sensitive, and low precision which means it has more false positives.
- (4) **LightGBM:** LightGBM achieved 78.19% accuracy, 0.6483 ROC-AUC, 0.2076 PR-AUC, and 0.2861 F1-score at a threshold of 0.15. The lower F1-score and recall indicate weaker performance compared to other lower accuracy scores for identifying imminent collapse events, although the accuracy is still relatively high.

3.7 Proposed Model: XGBoost

XGBoost was selected as the proposed model because it achieved the strongest overall performance among the five candidate models.

Table 3: Comparison of the proposed model with related cricket prediction studies.

Ref.	Task / Result	Difference
[8]	T20 win-probability estimation with improved Brier score and accuracy	No batting-collapse prediction
[9]	T20 match-outcome prediction with 76.9% accuracy	No short-horizon collapse label
[5]	Run-chase monitoring with 76–84% T20 accuracy	Mainly run-chase focused
[2]	ODI win-probability prediction with over 82% accuracy	ODI-based; match-outcome focused
[1]	ODI winner prediction with 72–76% accuracy	Predicts winner, not collapse risk
[12]	Batsman performance prediction with 72.84% accuracy	Individual-level prediction
[15]	Systematic review of machine learning applications in cricket	No predictive model developed
[13]	T20 score prediction with MAE = 1.73	Regression task, not collapse classification
[3]	ODI score prediction with 95% test accuracy and RMSE = 30.2	ODI score-focused
Proposed	T20 batting-collapse prediction with 80.38% accuracy and 0.7079 ROC-AUC	Predicts imminent collapse risk after each ball

The model was trained using one-hot-encoded categorical features, resulting in 15,908 input dimensions, and used GPU-based training with early stopping based on validation AUC. The best model converged at iteration 440, and the optimal decision threshold was set to 0.59 using validation-set F1-score.

On the independent test set, XGBoost achieved an accuracy of 80.38%, ROC-AUC of 0.7079, PR-AUC of 0.3148, and F1-score of 0.3429, with precision of 0.2911 and recall of 0.4172. These results were higher than those of the competing models, particularly in ROC-AUC, PR-AUC, and F1-score, which are important for the imbalanced collapse prediction task. The close validation and test performance also indicates that the model generalized well to unseen matches. Therefore, XGBoost was adopted as the proposed model for ball-level imminent batting collapse prediction, as shown in Figure 2.

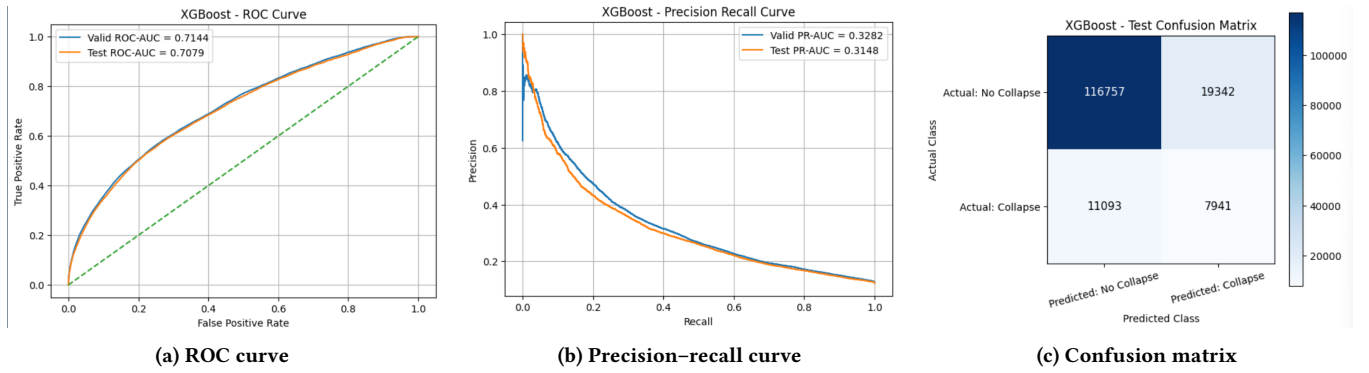


Figure 3: Evaluation of the proposed XGBoost model on the independent test set: (a) ROC curve, (b) precision–recall curve, and (c) confusion matrix.

4 Results and Discussion

Five candidate models—Logistic Regression, CatBoost, XGBoost, TCN, and LightGBM—were evaluated on an independent held-out test set for imminent batting collapse prediction. Performance was assessed using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and Brier Score. Given the class imbalance, where collapse cases represented only 12.41% of the dataset, F1-score, ROC-AUC, and PR-AUC were emphasized over accuracy. The comparative results are presented in Table 4.

Table 3 compares the proposed model with related cricket prediction studies. Existing studies mainly focus on final score prediction, match outcome prediction, win-probability estimation, or player performance analysis. In contrast, the proposed work focuses on short-horizon batting-collapse risk prediction after each legal delivery.

The detailed evaluation of the proposed XGBoost model is presented in Figure 3. Specifically, Figure 3a shows the ROC curve, which illustrates the model’s ability to distinguish between collapse and non-collapse states. Figure 3b presents the precision–recall curve, which is especially useful for evaluating performance under class imbalance. Figure 3c shows the confusion matrix, summarizing the distribution of correct and incorrect predictions on the independent test set.

Among the five models, XGBoost achieved the best overall performance, with the highest accuracy of 80.38%, ROC-AUC of 0.7079, PR-AUC of 0.3148, and F1-score of 0.3429. Compared with the Logistic Regression baseline, XGBoost improved ROC-AUC from 0.6506 to 0.7079 and PR-AUC from 0.2096 to 0.3148, indicating a stronger ability to rank collapse cases above non-collapse cases. It also had the best precision of all models analyzed which confirmed its reliability in identifying the high risk collapse state.

Logistic Regression gave a decent baseline with an F1 score of 0.2978 with an accuracy of 75.64%. However, its lower PR-AUC (0.2096) suggests that a linear model was not as effective in detecting the minority collapse class. While CatBoost had the second highest ROC-AUC of 0.6634 and enhanced PR-AUC compared to Logistic Regression, it didn’t perform as well as XGBoost on the F1 score. This indicates that categorical feature handling was beneficial, however it was not enough to outperform the proposed model.

The TCN model recorded the highest recall value, which was 0.4793, meaning that the sequential information from the previous 12 legal deliveries was good at identifying more true collapse cases. It had the lowest precision of 0.1852 and lowest F1-score of 0.2672 which indicates that it had more False Positives. This shows that a short-term sequence feature is not enough for reliable prediction of collapse without the use of more comprehensive match-state features. Although LightGBM’s accuracy is relatively high (78.19%), the F1, ROC-AUC and PR-AUC scores are lower than that of XGBoost, suggesting that LightGBM is not as effective as XGBoost in detecting the minority classes. The results suggest that ball-level match-state features can be useful cues for predicting imminent batting collapse overall. XGBoost was better at identifying important nonlinear patterns associated with the score state, the wicket pressure, the momentum lost and gained in the preceding two deliveries, the bowler and the batter, and the partnership. Thus, this was chosen to be the proposed classifier for ball-level batting-collapse risk prediction.

5 Limitations and Future Works

This study has some limitations. The collapse label was created with a fixed collapse window of 12 balls, and 2 wickets and 15 to 24 runs, this may not represent all collapse patterns. It was also a model that only included the ball-ball information available, rather than other factors like pitch condition, weather, player fitness, or form. Furthermore, the class distribution of the class was not balanced with collapse class. In addition, the detection of minority class was difficult because of imbalanced class distribution with collapse class. In the future, extensions are to use other collapse definitions, other prediction windows, and to incorporate other contextual variables like venue, pitch behaviour, player form and matchup history. Explainable AI techniques can also be integrated to help interpret predictions of collapse risk for coaches, analysts and real-time decision-support.

6 Conclusion

This study aimed to introduce a ball-by-ball machine learning method to forecast a potential collapse in batting, in the context of T20 cricket. The task was stated to be a binary classification

Table 4: Performance comparison of candidate models on the independent test set.

Model	Threshold	Accuracy	Precision	Recall	F1-score	ROC-AUC
TCN	0.52	67.75%	0.1852	0.4793	0.2672	0.6209
CatBoost	0.64	74.85%	0.2251	0.4295	0.2954	0.6634
Logistic Regression	0.57	75.64%	0.2304	0.4211	0.2978	0.6506
LightGBM	0.15	78.19%	0.2391	0.3562	0.2861	0.6483
XGBoost (Proposed)	0.59	80.38%	0.2911	0.4172	0.3429	0.7079

problem that must be handled with at least 15 balls of the next 12 balls of the next 12 balls are scored, and at least 2 wickets are lost by the batting side. Five models were tested on an independent test set using leakage-free match-state, scoring, wicket and recent-momentum features. XGBoost had the best overall performance, highlighting its effectiveness in detecting early signs of collapse risk. This work is different from the conventional cricket prediction studies which focus on the final score or the winner of the match, it offers a short-horizon warning approach that can aid in real-time cricket analysis and decision-making.

References

- [1] Yash Agrawal and Kundan Kandhway. 2024. Winner Prediction in an Ongoing One Day International Cricket Match. *Journal of Sports Analytics* 9, 4 (2024), 305–318. doi:10.3233/JSA-220735
- [2] Muhammad Asif and Ian G. McHale. 2016. In-Play Forecasting of Win Probability in One-Day International Cricket: A Dynamic Logistic Regression Model. *International Journal of Forecasting* 32, 1 (2016), 34–43. doi:10.1016/j.ijforecast.2015.02.005
- [3] Mazhar Javed Awan, Syed Arbaz Haider Gilani, Hamza Ramzan, Haitham Nobanee, Awais Yasin, Azlan Mohd Zain, and Rabia Javed. 2021. Cricket Match Analytics Using the Big Data Approach. *Electronics* 10, 19 (2021), 2350. doi:10.3390/electronics10192350
- [4] Rianka Bhattacharya, Paramjit S. Gill, and Tim B. Swartz. 2011. Duckworth–Lewis and Twenty20 Cricket. *Journal of the Operational Research Society* 62, 11 (2011), 1951–1957. doi:10.1057/jors.2010.175
- [5] Dipankar Bose and Soumyakanti Chakraborty. 2019. Managing In-Play Run Chases in Limited Overs Cricket Using Optimized CUSUM Charts. *Journal of Sports Analytics* 5, 4 (2019), 335–346. doi:10.3233/JSA-190342
- [6] Sanjay Chakraborty, Arnab Mondal, Aritra Bhattacharjee, Ankush Mallick, Riju Santra, Saikat Maity, and Lopamudra Dey. 2024. Cricket Data Analytics: Forecasting T20 Match Winners Through Machine Learning. *International Journal of Knowledge-Based and Intelligent Engineering Systems* 28, 1 (2024), 73–92. doi:10.3233/KES-230060
- [7] Cricsheet. 2026. Cricsheet: Ball-by-Ball Cricket Data. <https://cricsheet.org/>. Accessed: 2026-05-11.
- [8] Sonish Lamsal and David Kahle. 2024. In-Game Win Prediction Models for Cricket. In *Recent Advances in Next-Generation Data Science*, Henry Han (Ed.). Communications in Computer and Information Science, Vol. 2158. Springer, Cham, 148–168. doi:10.1007/978-3-031-67871-4_11
- [9] Abdul Majid, Qamruz Zaman, Ghazala Sahib, Soofia Iftikhar, Sundas Hussain, and Najma Salahuddin. 2025. Optimal Machine Learning Models for T20 Cricket: The Role of Dangerous Balls in Match Outcomes. *Metallurgical and Materials Engineering* 31, 4 (2025), 852–866. doi:10.63278/1522
- [10] Don K. Mallawa Arachchi, Ananda B. Manage, and Stephen M. Scariano. 2024. A Pressure Index for the Team Batting Second in T20I Cricket. *Journal of Sports Analytics* 10, 1 (2024), 123–135. doi:10.3233/JSA-240792
- [11] Michael J. Najdan, Matthew T. Robins, and Paul S. Glazier. 2014. Determinants of Success in English Domestic Twenty20 Cricket. *International Journal of Performance Analysis in Sport* 14, 1 (2014), 276–295. doi:10.1080/24748668.2014.11868721
- [12] Fawad Nasim, Muhammad Adnan Yousaf, Sohail Masood, Arfan Jaffar, and Muhammad Rashid. 2023. Data-Driven Probabilistic System for Batsman Performance Prediction in a Cricket Match. *Intelligent Automation & Soft Computing* 36, 3 (2023), 2865–2877. doi:10.32604/iasc.2023.034258
- [13] Akhil Tyagi, Amandeep Kaur, Aryan Kamboj, Chayandeep Chaulia, Gandharv Mohan, and Manpreet Singh. 2024. XGBoosting Cricket: Enhancing Predictive Modeling for Twenty20 Match Results Using Machine Learning and Statistical Techniques. *SN Computer Science* 5, 8 (2024), 1036. doi:10.1007/s42979-024-03385-0
- [14] Sasank Viswanadha, Kaustubh Sivalenka, Madan Gopal Jhavar, and Vikram Pudi. 2017. Dynamic Winner Prediction in Twenty20 Cricket: Based on Relative Team Strengths. In *Proceedings of the Machine Learning and Sports Analytics Workshop*. Leuven, Belgium. https://dtai.cs.kuleuven.be/events/MLSA17/papers/MLSA17_paper_5.pdf
- [15] Indika Wickramasinghe. 2022. Applications of Machine Learning in Cricket: A Systematic Review. *Machine Learning with Applications* 10 (2022), 100435. doi:10.1016/j.mlwa.2022.100435